

POS System

User Guide

Restaurant Management Platform
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Prepared for: FnB Platform Stakeholders & User Community

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Table of Contents

Executive Summary

System Architecture

User Journey Maps

1. System Overview

2. Accessing the System

3. Demo Credentials

4. User Roles & Permissions

5. Navigation & Layout

6. Dashboard

7. Order Management

8. Menu Management

9. Inventory Management

10. Restaurant & Building Management

11. User Management

12. Module Access Control

13. QR Code Scanner

14. Delivery Confirmation

15. Module Management (Super Admin)

Glossary

Appendix

Executive Summary

Product Overview

The POS System is an enterprise-grade restaurant management platform designed for multi-location food and beverage operations in the Indian market. It provides end-to-end digital management of restaurant operations including menu display, order processing, inventory tracking, user administration, and delivery confirmation. The platform is built on a modern, scalable architecture using React, Node.js, and PostgreSQL, and is accessible from any modern web browser without software installation.

Key Capabilities

Capability	Description	Business Impact
Multi-Location Management	Manage restaurants across buildings	Centralised control, consistent brand experience
Real-Time Order Processing	End-to-end order lifecycle with live updates	Faster service, reduced errors
Inventory Automation	Auto-deduction on order completion	Accurate stock levels, reduced waste
Role-Based Security	5-tier access control hierarchy	Data security, regulatory compliance
QR Code Integration	Camera-based order lookup and delivery	Instant identification, zero typing errors
Comprehensive Reporting	Real-time KPIs and analytics dashboards	Data-driven decision making

Supported User Roles

Role	Level	Scope	Primary Responsibilities
Super Admin	100	System-wide	Configuration, user management, module control
Building Manager	70	Assigned building	Building oversight, restaurant management
Restaurant Manager	50	Assigned restaurant	Daily operations, orders, inventory, staff
Chef	40	Assigned restaurant	Order preparation, inventory checks, wastage
Customer	10	Self	Menu browsing, ordering, order history

Business Benefits

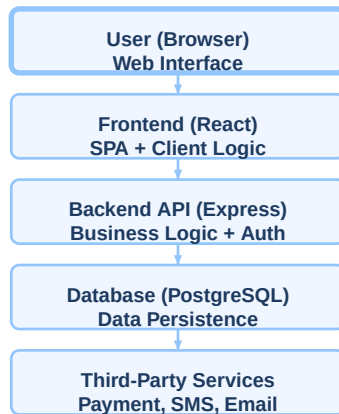
The POS System delivers measurable operational improvements: elimination of manual order errors through digital workflows, reduction of stock wastage via automated inventory tracking, faster customer service through QR-based identification, improved staff productivity with role-optimised interfaces, and enhanced management visibility through real-time dashboards and reports.

System Architecture

High-Level Architecture

The POS System follows a three-tier web architecture comprising a presentation layer (React SPA), an application layer (Node.js/Express API), and a data layer (PostgreSQL). The frontend communicates with the backend over HTTPS with JSON payloads, while real-time updates are delivered through Socket.IO WebSocket connections. All services are containerised using Docker and deployed on Railway with automated CI/CD via GitHub Actions.

System Architecture Flow



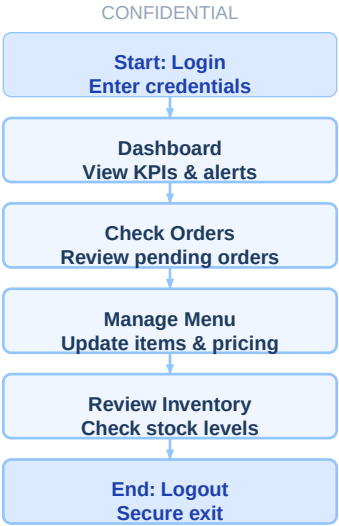
The frontend layer renders the user interface, handles client-side routing, and manages local state. The backend layer processes all business logic, enforces security rules, and orchestrates data operations. The database layer stores all transactional and master data. Third-party integrations handle payment processing, SMS notifications, and email communications.

User Journey Maps

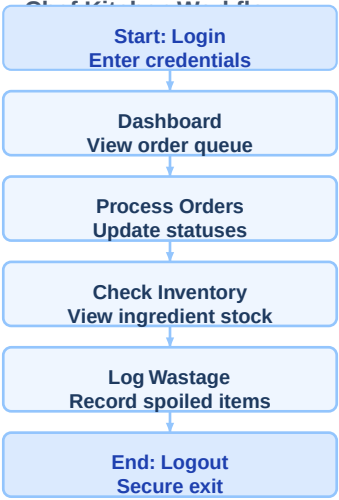
The following journey maps illustrate how different roles interact with the system during a typical work session. These maps help new users understand the application flow quickly.

Restaurant Manager Journey

Manager Daily Workflow



Chef Journey



Administrator Journey
Admin Configuration Workflow



1. System Overview

Overview

The POS System is a comprehensive restaurant management platform purpose-built for the Indian food and beverage market. It serves as the central operating system for multi-location restaurant chains, integrating menu management, order processing, inventory tracking, user administration, and delivery confirmation into a single unified platform. The system bridges the gap between front-of-house operations and back-of-house management, enabling seamless data flow across all functional areas.

Built on a modern technology stack — React 18 on the frontend with Node.js, Express, and PostgreSQL on the backend — the system delivers a responsive, real-time experience accessible from any modern web browser. It supports complex organisational hierarchies where multiple restaurant outlets operate within larger building complexes, with role-based security ensuring that each user sees only the data and features appropriate to their responsibilities.

Capability	Description	Business Benefit
Multi-Location Support	Manage outlets across buildings	Centralised control, consistent operations
Role-Based Security	5-tier access hierarchy	Data security, operational compliance
Real-Time Updates	Socket.IO push notifications	Immediate visibility, faster response
Inventory Automation	Auto-deduction on order completion	Accurate stock, reduced waste
QR Integration	Camera-based order lookup	Faster service, fewer errors
Web-Based Access	No installation required	Lower IT overhead, device flexibility

Technology Stack

Layer	Technology	Purpose
Frontend	React 18, Vite 5, Tailwind CSS 3	User interface and client-side logic
Backend	Node.js, Express, Prisma ORM	API server and business logic
Database	PostgreSQL	Data persistence and transactions
Real-Time	Socket.IO	Live order status updates
Authentication	JWT + bcrypt	Stateless authentication
Deployment	Railway, Docker, GitHub Actions	Hosting and CI/CD pipeline

2. Accessing the System

Overview

The POS System is entirely web-based and requires no software installation. Users access the platform through any modern web browser on desktop computers, laptops, tablets, or mobile phones. The login process authenticates the user, determines their role-based permissions, and loads a personalised dashboard tailored to their responsibilities within the organisation.

Business Objective

Provide a secure, frictionless entry point for all users regardless of device or location, ensuring that the correct user gains access to the correct set of features based on their organisational role.

User Roles

Role	Access Level	Actions Allowed
All Registered Users	Full	Login, access permitted modules based on role

Navigation Path

Open a supported web browser. Navigate to: <https://fnb-mvp.up.railway.app>. Supported browsers: Chrome 90+, Edge 90+, Firefox 90+, Safari 15+.

Detailed Workflow

When a user navigates to the frontend URL, the system serves the React single-page application. The login page renders with username and password fields. Upon submission, the frontend sends a POST request to `/api/auth/login` with the provided credentials. The backend retrieves the user record from PostgreSQL, compares the password against the stored bcrypt hash, and if validated, generates a JSON Web Token (JWT) containing the user's identity, role, building ID, and restaurant ID.

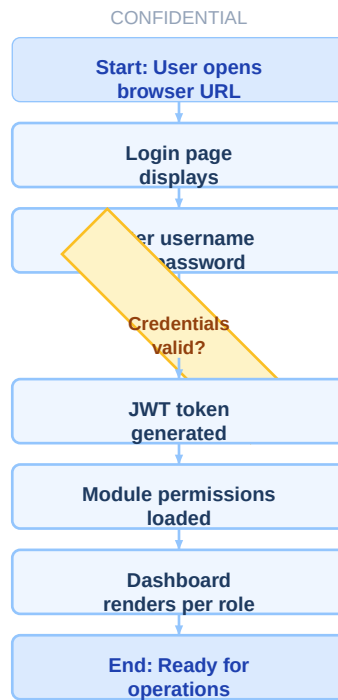
The JWT is returned to the frontend and stored in the browser's localStorage. The frontend then immediately calls `GET /api/modules/my-access` to load the user's module permissions. Based on the returned allowed modules, the sidebar navigation is filtered, and the role-appropriate dashboard is rendered. The user can then navigate to any permitted module.

Figure 1 – Login Screen

The login page is the single entry point for all system users.

- Username and password input fields with validation
- Login button with loading indicator
- Error message display for invalid credentials

Login Process



Example Scenario

A newly hired Restaurant Manager on their first day opens Chrome, enters the frontend URL, and logs in with provided credentials. The system verifies their identity, loads their module permissions, and presents the manager dashboard. The sidebar shows permitted modules: Dashboard, Orders, Inventory, Menu, Users, and Delivery. The manager is ready to begin daily operations immediately.

Business Rules

- JWT tokens expire after 7 days, requiring re-authentication upon expiry.
- Passwords are stored as bcrypt hashes; plain-text passwords are never persisted.
- Session state is maintained in browser localStorage; clearing browser data logs the user out.
- The QR scanner feature requires browser camera permissions to be granted by the user.
- After multiple failed login attempts, a security delay is imposed to prevent brute force attacks.

Expected Outcome

Upon successful authentication, the user is presented with a role-specific dashboard containing relevant KPIs, quick-action shortcuts, and a sidebar filtered to show only the modules they are permitted to access.

3. Demo Credentials

Overview

The system includes pre-seeded demonstration accounts for all five user roles. These accounts are created automatically when the database seed script runs and come pre-configured with sample buildings, restaurants, menu items, orders, and inventory records. Demo accounts enable prospective clients, trainees, and implementation consultants to explore the full feature set without setting up production data.

Role	Username	Password	Access Level
Super Admin	Superadmin	Admin12345	Full system access
Building Manager	bldmgr1	bldmgr123	Building oversight
Restaurant Manager	restmgr1	restmgr123	Outlet operations
Chef	chef1	chef123	Order processing
Customer	customer1	customer123	Menu browsing & ordering

WARN: Demo accounts should never be used in production environments due to shared passwords. Production deployments must create unique credentials for each individual user.

4. User Roles & Permissions

Overview

The POS System implements a hierarchical role model with five distinct access levels. Each role inherits permissions from roles below it, and users can only manage accounts with roles strictly lower than their own. This structure mirrors typical restaurant organisational hierarchies and ensures appropriate access control at every level. Role enforcement happens at multiple layers: the frontend hides inaccessible UI elements, the sidebar filters navigation links, and the backend API independently verifies permissions on every request. This multi-layered approach ensures security even if a user attempts to directly navigate to a restricted URL.

Role (Level)	Scope	Key Permissions
Super Admin (100)	Entire system	Full CRUD on all entities, module configuration, role switching
Building Manager (70)	Assigned building	Manage restaurants, create RM/Chef/Customer users
Restaurant Manager (50)	Assigned restaurant	Menu, orders, inventory, users (Chef/Customer)
Chef (40)	Assigned restaurant	View/update orders, view inventory, log wastage
Customer (10)	Self	Browse menus, place orders, view order history

NOTE: Super Admin bypasses all role-level checks and has unrestricted access across all buildings and restaurants. This is by design to enable system-wide administration.

5. Navigation & Layout

Overview

The user interface follows a consistent three-component layout: a Top Bar providing context and user controls, a Sidebar (desktop) or Bottom Navigation (mobile) listing available modules, and a main content area. Navigation links are dynamically filtered by the user's role and enabled modules.

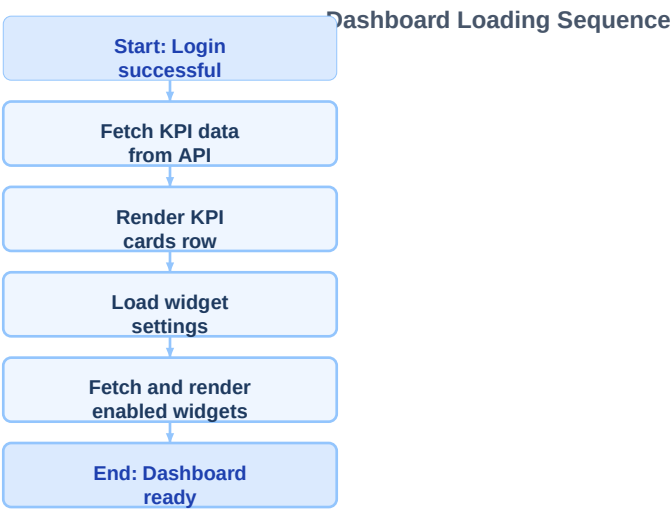
Component	Visibility	Function
Top Bar	All authenticated users	Logo, context display, notifications, theme toggle, profile
Sidebar	Desktop screens	Role-filtered navigation links with active highlighting
Bottom Navigation	Mobile screens	Up to 6 frequently used tabs, filtered by role
Hamburger Menu	Mobile screens	Full module list overlay for less frequent features

6. Dashboard

Overview

The Dashboard serves as the central command centre and default landing page after login. It provides an at-a-glance view of business performance through KPI cards, quick-action shortcuts, and a customisable widget system with eight panel types. Super Administrators see additional Building Reports with cross-location aggregated data.

Role	Data Scope	Additional Widgets
Super Admin	Organisation-wide	Building Reports (cross-location comparison)
Building Manager	Assigned building	Standard KPI widgets
Restaurant Manager	Assigned restaurant	Outlet-specific KPIs
Chef	Assigned restaurant	Order-focused KPIs
Customer	Self	Order history, favourite items



7. Order Management

Overview

Order Management handles the complete order lifecycle from placement through delivery across five stages: PENDING_PAYMENT, PAID, PREPARING, COMPLETED, and DELIVERED. Real-time updates via Socket.IO ensure all staff see changes instantly. When an order reaches COMPLETED, the system automatically deducts corresponding ingredient quantities from inventory if recipe mappings exist.

Role	Access Level	Actions Allowed
Super Admin	Read-only	Monitor orders across all locations
Building Manager	Read-only	Monitor orders in assigned building
Restaurant Manager	Full	View, filter, update status, manage all orders
Chef	Operational	View orders, update status (PAID to COMPLETED)
Customer	Self	Place orders, view personal order history

Order Lifecycle

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Business Rules

- Order statuses must progress sequentially and cannot be reversed to a previous status.
 - Inventory deduction triggers only on COMPLETED status when recipe mappings exist.
 - Order codes are unique 6-character alphanumeric strings generated at order creation.
 - Socket.IO broadcasts all status changes to connected clients in real-time.
-

8. Menu Management

Overview

The Menu Management module enables restaurants to create, edit, and maintain digital menu offerings. Each item includes a name, description, price in INR, food category (VEG, NON_VEG, or VEGAN with colour-coded indicators), an optional image, and an active/inactive toggle. Changes take effect immediately in the customer-facing ordering interface.

Role	Access Level	Actions Allowed
Super Admin	Full	Create, edit, delete menu items across all restaurants
Restaurant Manager	Full	Create, edit, delete menu items for their restaurant
Building Manager	Read-only	View menus across the building
Chef	Read-only	View menu items and ingredient mappings
Customer	Browse	View active menu items during ordering

Business Rules

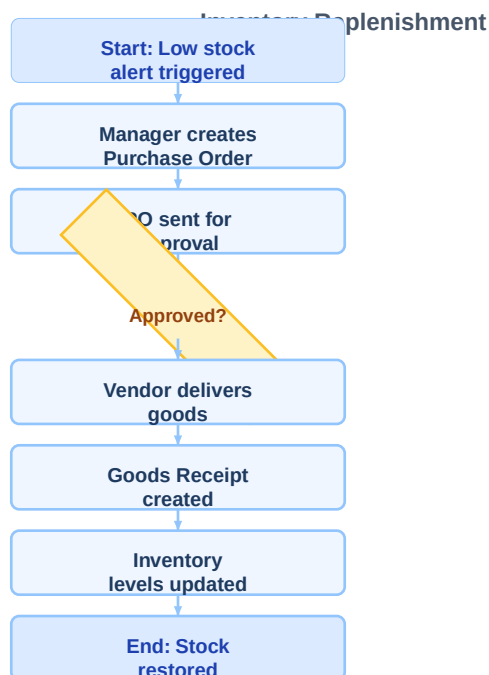
- Item name and price are required fields; description and image are optional.
 - Prices are in Indian Rupees as whole numbers without decimal places.
 - Only active items are visible to customers; inactive items retain database records.
 - Each restaurant maintains its own independent menu.
-

9. Inventory Management

Overview

The Inventory Management module provides comprehensive control over restaurant stock with 12 integrated sections covering the complete inventory lifecycle. The auto-deduction feature, powered by recipe mappings, eliminates manual stock adjustments and provides real-time inventory accuracy. The structured procurement workflow ensures optimal stock levels through systematic ordering and approval controls.

Role	Access Level	Actions Allowed
Super Admin	Full	All inventory functions across all restaurants
Building Manager	High	View and manage inventory across building restaurants
Restaurant Manager	Full	Complete inventory control for their restaurant
Chef	Limited	View inventory, log wastage



Business Rules

- Auto-deduction only triggers on **COMPLETED** order status with existing recipe mappings.
 - Transfers and Adjustments require managerial approval before execution.
 - All stock changes create an **InventoryMovement** audit record for complete traceability.
-

10. Restaurant & Building Management

Overview

Buildings represent physical locations such as shopping malls or complexes. Restaurants are operational outlets within buildings. This hierarchy scopes all other modules: menus, orders, inventory, and users belong to a specific restaurant within a specific building. A Building Manager oversees all restaurants in their building; each Restaurant Manager handles their own outlet.

Role	Buildings	Restaurants
Super Admin	Create, edit, delete all	Create, edit, delete all
Building Manager	View assigned building	View, edit in assigned building
Restaurant Manager	Read-only	View, edit own restaurant

Business Rules

- Each Building can contain multiple Restaurants; each Restaurant belongs to exactly one Building.
- Deactivating a Building hides all its Restaurants from active views.
- Buildings and Restaurants with associated orders or users cannot be permanently deleted.

11. User Management

Overview

The User Management module provides centralised control over system access. Administrators can create, edit, and delete user accounts, assign roles with appropriate permissions, and configure module-level access toggles. User management is distributed according to the role hierarchy, allowing each management level to handle their own staffing needs.

Role	User Scope	Manageable Roles
Super Admin	All users (system-wide)	All roles (100, 70, 50, 40, 10)
Building Manager	Users in assigned building	RM (50), Chef (40), Customer (10)
Restaurant Manager	Users in assigned restaurant	Chef (40), Customer (10)

Business Rules

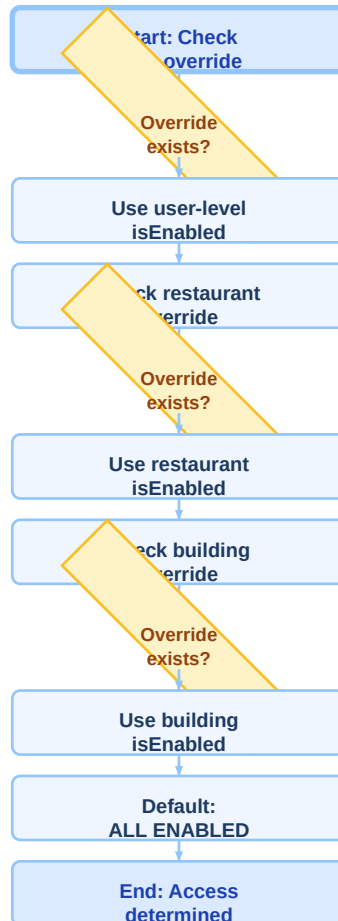
- Users can only manage accounts with roles strictly below their own level.
- Self-deletion is prevented by the system.
- Passwords are stored as bcrypt hashes; plain text is never persisted.

12. Module Access Control

Overview

Module Access Control enables Super Administrators to fine-tune which modules each user can access, overriding default role-based permissions. The resolution follows a three-tier priority system: User-level overrides take highest precedence, followed by Restaurant-level, then Building-level, with a default fallback of all modules enabled.

Module Resolution Priority



Business Rules

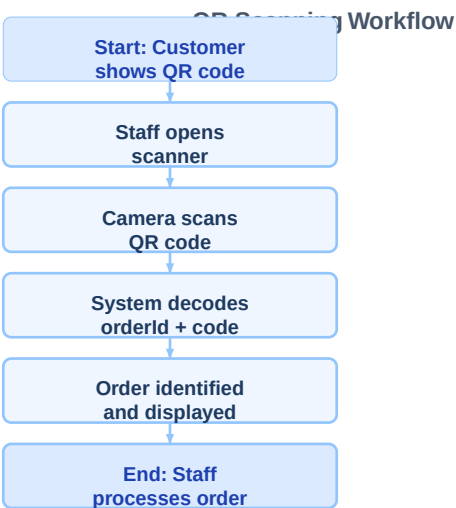
- Resolution priority: User-level > Restaurant-level > Building-level > Default (all enabled).
 - Super Admin bypasses all override checks and sees every module.
 - Overrides use an upsert pattern: create new or update existing as needed.
-

13. QR Code Scanner

Overview

The QR Code Scanner enables staff to instantly identify orders using the customer's digital receipt. Integrated into both the Orders page and Delivery Confirmation page, the scanner reads QR codes encoding { orderId, orderCode } for fast, error-free order lookup that eliminates manual data entry errors.

Role	Access Level	Actions Allowed
Restaurant Manager	Full	Scan orders on Orders page and Delivery Confirmation
Chef	Operational	Scan orders on Orders page



Business Rules

- Camera permission must be granted by the browser for the scanner to function.

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- Only PAID and COMPLETED orders display QR codes for scanning.

14. Delivery Confirmation

Overview

Delivery Confirmation is the terminal step in the order lifecycle. Staff formally record that an order has been handed over to the customer, updating the status to DELIVERED. The feature supports QR code scanning for instant identification and manual order code entry as a fallback.

Role	Access Level	Actions Allowed
Restaurant Manager	Full	Confirm delivery for all orders in their restaurant
Chef	Operational	Confirm delivery for prepared orders



Business Rules

- Confirmation only works for orders currently in COMPLETED status.

- Delivery confirmation is irreversible; no undo is available.

15. Module Management (Super Admin)

Overview

Module Management provides a comprehensive interface for Super Administrators to configure system-wide module overrides at the Building, Restaurant, or User level. This page offers a centralised view of all existing overrides and enables bulk permission changes that cascade to all affected users.

Role	Access Level	Actions Allowed
Super Admin	Full	View all modules, configure overrides at all levels
All Other Roles	No Access	Module Management page is not visible

Business Rules

- Only Super Administrators can access this page and its API endpoints.
- Resolution priority: User-level > Restaurant-level > Building-level > Default (enabled).
- Super Admin bypasses all override checks regardless of settings.

Glossary

Term	Definition
API	Application Programming Interface — a set of protocols for building and integrating application software
CRUD	Create, Read, Update, Delete — the four basic operations of persistent storage
GRN	Goods Receipt Note — a document acknowledging receipt of goods from a vendor
JWT	JSON Web Token — a compact, URL-safe token format used for authentication
KDS	Kitchen Display System — a digital display showing incoming orders to kitchen staff
KPI	Key Performance Indicator — a measurable value that demonstrates how effectively objectives are achieved
ORM	Object-Relational Mapping — a technique for converting data between type systems in OOP languages
PO	Purchase Order — a commercial document issued by a buyer to a seller
POS	Point of Sale — the place where a retail transaction is completed
RBAC	Role-Based Access Control — an approach to restricting system access to authorised users
SKU	Stock Keeping Unit — a unique identifier for each product or item in inventory
SPA	Single-Page Application — a web application that loads a single HTML page and dynamically updates content
UAT	User Acceptance Testing — the final phase of testing before system deployment

Appendix

Version History

Version	Date	Author	Changes
1.0	June 2026	POS System Team	Initial release

Known Limitations

- The dashboard does not auto-refresh; users must reload the page to view updated KPI data.
- QR code scanning requires an active camera and may not function on all mobile browsers.
- The system currently supports Indian Rupees (INR) as the sole currency.

Future Enhancements

- Real-time dashboard auto-refresh via WebSocket push updates.
- Multi-currency support for international restaurant chains.
- Advanced analytics with predictive inventory and demand forecasting.
- Offline mode for order processing during network interruptions.
- Mobile native applications for iOS and Android platforms.

Support Contact

For technical support, feature requests, or issue reporting, contact the POS System team through the project repository at <https://github.com/PugazhendhiS-alt/FNB> or raise a ticket through the organisation's IT support portal.